A translation counterexample to TLMC 00000002685

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Abstract

The stated orbit-closure dichotomy fails for a single polynomial automorphism of the affine plane. Translation in one coordinate has constant Jacobian determinant one, no preperiodic points, and an orbit contained in a proper closed line. We give a characteristic-zero argument and a kernel-checked Lean formalization on the rational affine plane. The formalization uses actual polynomial zero sets, not an axiomatized closure operation.

1 Statement and interpretation

TLMC 00000002685 proposes that an orbit closure for a multidimensional endomorphism semigroup is either the preperiodic point set or Zariski dense, with emptiness of a multiplicative zero set of Jacobian determinants as a criterion, computably verifiable for at most two maps [1]. The text does not restrict the base field, impose degree at least two, or exclude invariant proper subvarieties.

We use the standard interpretation of endomorphisms as algebraic self-maps of an affine variety. For a single generator F, a point p is preperiodic if $F^m(p) = F^n(p)$ for some integers $0 \le m < n$. This agrees with finiteness of its forward orbit. Our example also refutes the weaker reading that the closure merely consists of preperiodic points. It is not a claim about endomorphisms required to preserve an algebraic-group identity; translation would not be a group homomorphism under that different interpretation.

2 Counterexample

Proposition 1. *Let k be any field of characteristic zero. On $\mathbb{A}^2_k$, the one-generator semigroup generated by*

$$F(x, y) = (x + 1, y)$$

has an orbit closure which is neither the preperiodic point set nor Zariski dense, although its Jacobian determinant and all finite products of that determinant are everywhere nonzero.

Proof. The map F is a polynomial automorphism, with inverse $(x, y) \mapsto (x - 1, y)$. Induction gives

$$F^n(a, b) = (a + n \cdot 1_k, b) \qquad (n \ge 0).$$

If $F^m(a, b) = F^n(a, b)$, then $(n - m) \cdot 1_k = 0$. Characteristic zero implies $m = n$. Thus there are no preperiodic points, over k or over any extension field of k.

Take $p = (0, 0)$. Its forward orbit is

$$\mathcal{O}(p) = \{(n \cdot 1_k, 0) : n \ge 0\}.$$

Its Zariski closure C contains p, and hence is nonempty. The polynomial Y vanishes on the whole orbit, so $C \subseteq V(Y)$. This is a proper closed subset, since $(0, 1) \notin V(Y)$. Therefore C

is not dense, and its nonemptiness prevents it from equalling the empty preperiodic point set. Indeed, it does not even consist of preperiodic points.

Finally,

$$DF = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \det DF = 1.$$

Every finite product of this determinant is 1, so its zero set is empty. The same holds for determinants of iterates, and for compositions or pullbacks of these constant determinant functions. Thus the proposed Jacobian condition does not rescue the dichotomy. One generator is already within the stated limit of at most two maps. □

Only the containment $C \subseteq V(Y)$ is needed. No identification of C with the entire line is used. If a semigroup convention excludes the identity iterate, the orbit starts at $(1, 0)$ instead; it is still nonempty, contained in the same proper closed line, and contains no preperiodic point. The same counterexample works on $\mathbb{A}^d_k$ for every $d \geq 2$ by translating the first coordinate and fixing the others.

3 What is formally verified

The accompanying `Main.lean` is self-contained apart from Lean's standard distribution. It imports `Std` and rational arithmetic lemmas, and uses Lean 4.31.0. Its domain is exactly $\mathbb{Q} \times \mathbb{Q}$. It does not formalize arbitrary fields, complex points, schemes, or the extension-of-scalars argument above. The preceding characteristic-zero proof is an ordinary mathematical proof; only its rational-point instance is kernel-checked here.

A polynomial is represented by a finite expression generated by rational constants, variables X, Y, addition, and multiplication. These expressions represent all elements of $\mathbb{Q}[X, Y]$; repeated syntactic representatives do not alter common zero sets. For a set S of rational points, the formal definition is exactly

$$\overline{S} = \{q : \ \forall f, \ (\forall r \in S, \ f(r) = 0) \Rightarrow f(q) = 0\}.$$

This is the Zariski closure for rational-coefficient polynomials on rational points. Quantification is over all polynomial expressions, not a selected list of equations. Density means that this closure contains every point. The coordinate formulas of F are polynomial expressions; the code also checks the iterate formula, the composition law for iterates, and a polynomial representation of every iterate.

The final theorem `TLMC2685.counterexample` combines emptiness of the multiplicative Jacobian zero set with failure of both alternatives. Its second alternative explicitly asks for density of the orbit *closure*, not merely of the original orbit. A separate theorem shows the closure is not even contained in the preperiodic set. The Jacobian is computed from recursively defined formal partial derivatives of the actual coordinate polynomials.

The axiom audit reports only Lean's standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`. There are no additional axioms, proof holes, `sorry`, or `native_decide` steps. The formal result deliberately makes no claim that the orbit closure equals the line, or that the conjecture's unspecified computability criterion has been independently defined in full generality. Since the claimed dichotomy already fails, no analysis of that algorithmic clause is needed for the counterexample.

References

[1] The Last Math Competition, Conjecture 00000002685, repository source, accessed 12 October 2026. File blob SHA: `8d6a6cbadf48e960123f588074f14019237a7581`.